

Natural Computing

LMU Munich
summer term 2025

Thomas Gabor



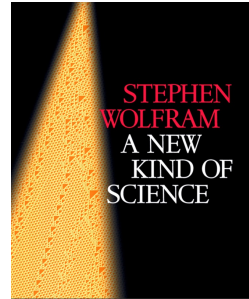
Who knows Conway's Game of Life?



Cellular Automata

Wolfram.
A New Kind of Science.
Online, 2002.

www.wolframscience.com/nks/



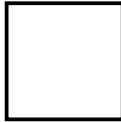
Shiffman.
The Nature of Code.
Online, 2012.

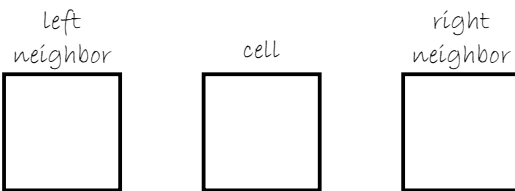
natureofcode.com/book/



What was an automaton again?

cell





Definition 1 (cellular automaton). Let $G = (V, E)$ be a graph with vertices V and edges $E \subseteq V \times V$. Let $neighborhood : V \rightarrow V^{d+1}$ for some neighborhood degree $d \in \mathbb{N}$ be a function that returns an ordered vector of neighbors of a given node v , always including v itself. A state $x \in \mathcal{X}$ is a mapping of vertices to the values $\{0, 1\}$, i.e., the state space $\mathcal{X}$ is given via $\mathcal{X} = (V \rightarrow \{0, 1\})$. Let x_t be a state that exists at time step $t \in \mathbb{N}$. The evolution of a state x_t to its subsequent state x_{t+1} is given deterministically via a function $f : \{0, 1\}^{d+1} \rightarrow \{0, 1\}$ so that

$$x_{t+1}(v) = f(x_t(u_1), \dots, x_t(u_{d+1}))$$

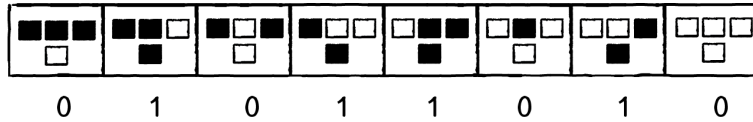
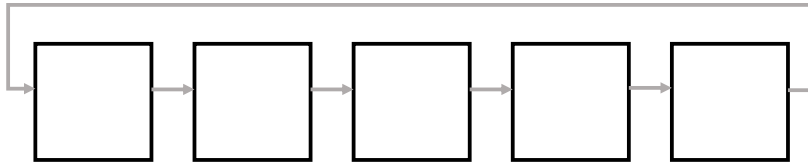
where $\vec{u} = \langle u_1, \dots, u_{d+1} \rangle = neighborhood(v)$.

A tuple (G, f, x_0) is called a cellular automaton with initial state $x_0 \in \mathcal{X}$.

Definition 2 (1D cellular automaton). A cellular automaton (G, f, x_0) is called a 1D cellular automaton iff all vertices have exactly one incoming edge (“left neighbor”) and one outgoing edge (“right neighbor”) and $neighbors(v) = \langle u, v, w \rangle$ where $(u, v) \in E$ and $(v, w) \in E$ and thus $d = 2$. Subsequently, we usually write

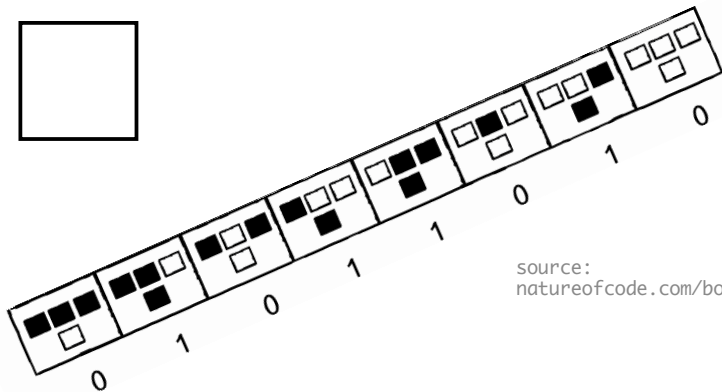
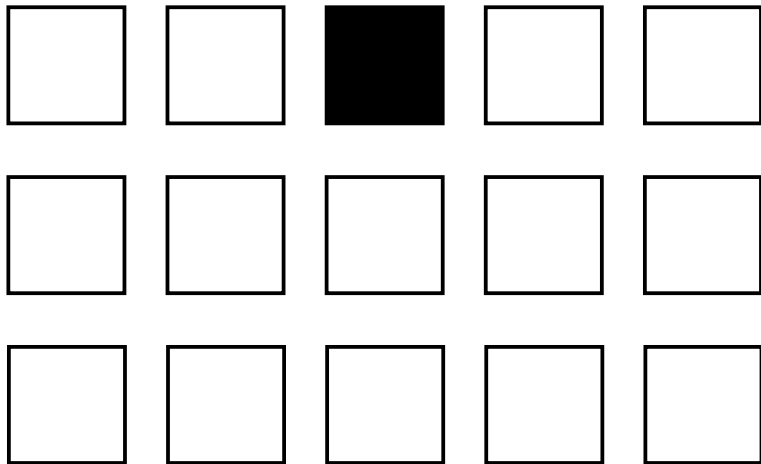
$$x_{t+1}(v) = f(x_t(u), x_t(v), x_t(w))$$

for the evolution where u is the left neighbor of v and w is the right neighbor of v .



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Why?

Experiment Time!

Experiment Results

your
result

my
result

Experiment 1:

average

Experiment 2:

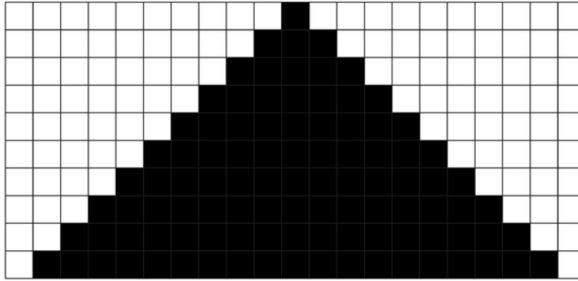
most frequent – whisper only

Experiment 3:

highest – whisper only

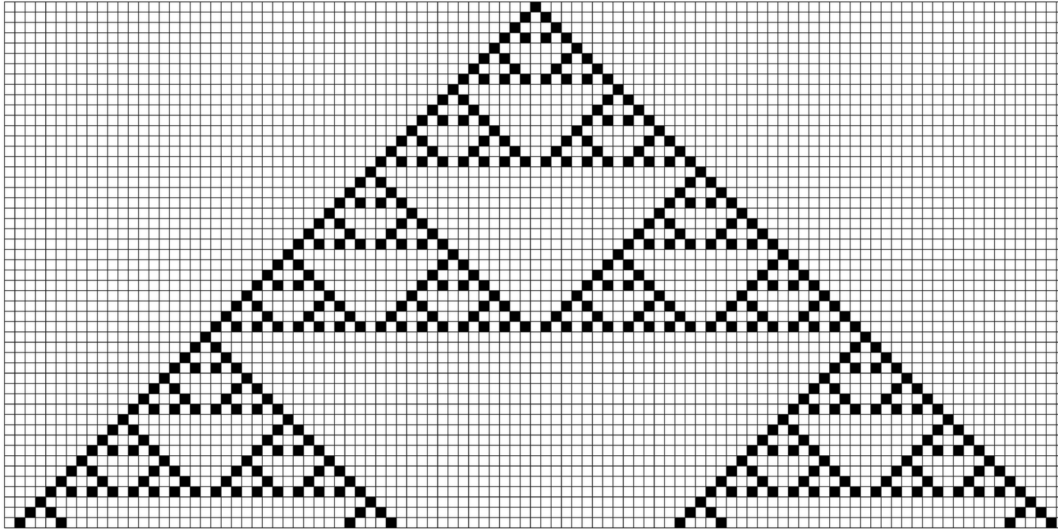
Is local information sufficient
for meaningful computation?

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wolframscience.com/nks

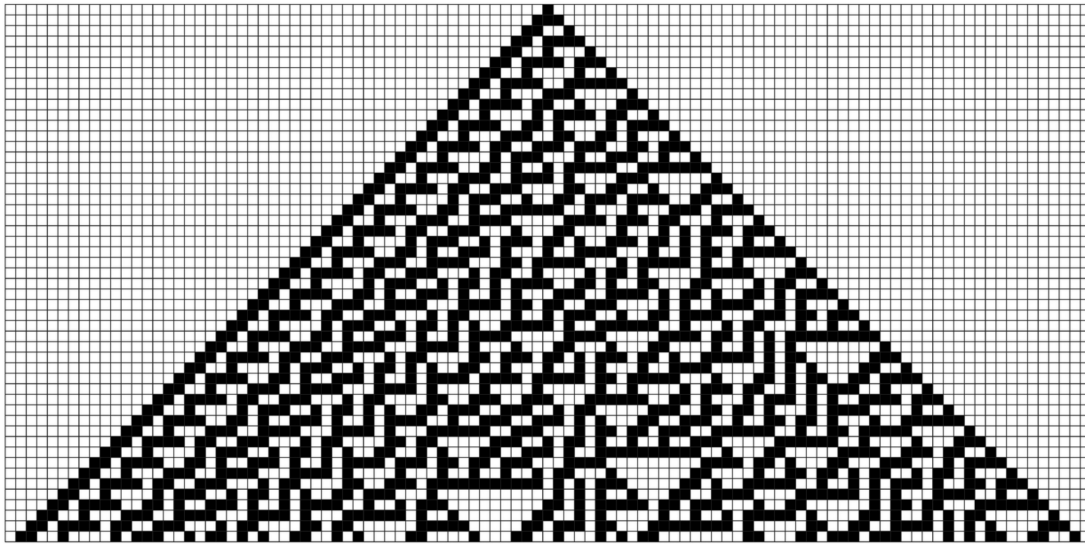


gives one of the possible combinations of colors for a cell and its immediate neighbors. The bottom row then specifies what color the center cell should be on the next step in each of these cases. In the numbering scheme described in Chapter 3, this is cellular automaton rule 254.

A representation of the rule for the cellular automaton shown above. The top row in each box



A cellular automaton that produces an intricate nested pattern. The rule in this case is that a cell should be black whenever one or the other, but not both, of its neighbors were black on the step before. Even though the rule is very simple, the picture shows that the overall pattern obtained over the course of 50 steps starting from a single black cell is not so simple. The particular rule used here can be described by the formula $a'_i = \text{Mod}[a_{i-1} + a_{i+1}, 2]$. In the numbering scheme of Chapter 3, it is cellular automaton rule 90.

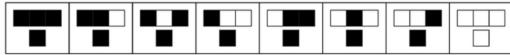


A cellular automaton with a simple rule that generates a pattern which seems in many respects random. The rule used is of the same type as in the previous examples, and the cellular automaton is again started from a single black cell. But now the pattern that is obtained is highly complex, and shows almost no overall regularity. This picture is our first example of the fundamental phenomenon that even with simple underlying rules and simple initial conditions, it is possible to produce behavior of great complexity. In the numbering scheme of Chapter 3, the cellular automaton shown here is rule 30.

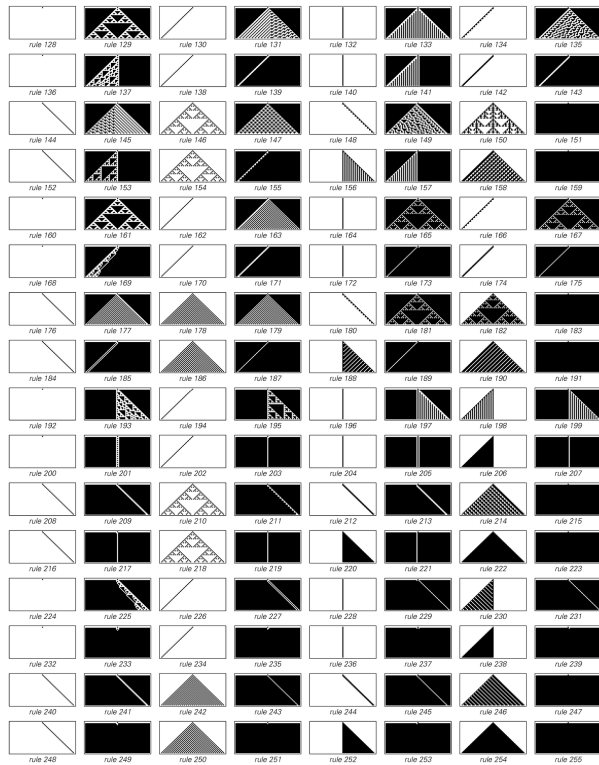
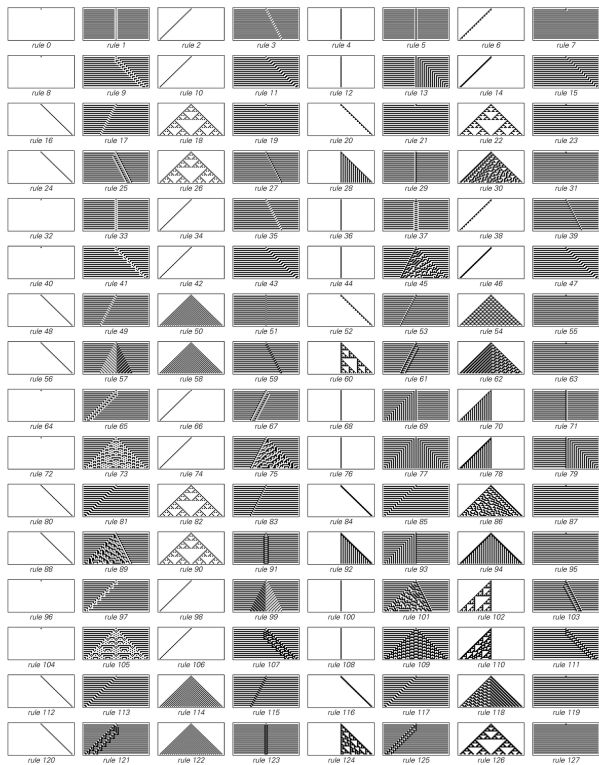
On Names for Rules



A cellular automaton that produces an intricate nested pattern. The rule in this case is that a cell should be black whenever one or the other, but not both, of its neighbors were black on the step before. Even though the rule is very simple, the picture shows that the overall pattern obtained over the course of 50 steps starting from a single black cell is not so simple. The particular rule used here can be described by the formula $a_i' = \text{Mod}[a_{i-1} + a_{i+1}, 2]$. In the numbering scheme of Chapter 3, it is cellular automaton rule 90.



A representation of the rule for the cellular automaton shown above. The top row in each box gives one of the possible combinations of colors for a cell and its immediate neighbors. The bottom row then specifies what color the center cell should be on the next step in each of these cases. In the numbering scheme described in Chapter 3, this is cellular automaton rule 254.



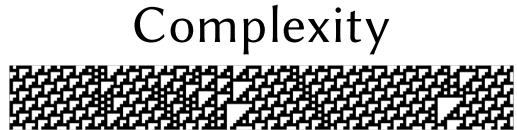
Classes of Cellular Automata



Uniformity



Repetition



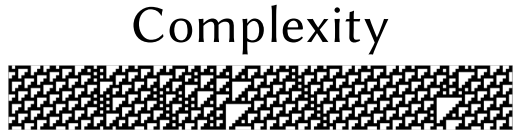
Classes of Cellular Automata



Uniformity



Repetition





source: commons.wikimedia.org/wiki/File:Textile_cone.JPG